Blood Harmane, Blood Lead, and Severity of Hand Tremor: Evidence of Additive Effects

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Background Tremor is a widespread phenomenon in human populations. Environmental factors are likely to play an etiological role. Harmane (1-methyl-9H-pyrido[3,4-β]indole) is a potent tremor-producing β-carboline alkaloid. Lead is another tremor-producing neurotoxicant. The effects of harmane and lead with respect to tremor have been studied in isolation. Objectives We tested the hypothesis that tremor would be particularly severe among individuals who had high blood concentrations of both of these toxicants. Methods Blood concentrations of harmane and lead were each quantified in 257 individuals (106 essential tremor cases and 151 controls) enrolled in an environmental epidemiological study. Total tremor score (range = 0 – 36) was a clinical measure of tremor severity. Results The total tremor score ranged from 0 – 36, indicating that a full spectrum of tremor severities was captured in our sample. Blood harmane concentration correlated with total tremor score ($p = 0.007$), as did blood lead concentration ($p = 0.045$). The total tremor score was lowest in participants with both low blood harmane and lead concentrations (8.4 ± 8.2), intermediate in participants with high concentrations of either toxicant (10.5 ± 9.8), and highest in participants with high concentrations of both toxicants (13.7 ± 10.4)($p = 0.01$). Conclusions Blood harmane and lead concentrations separately correlated with total tremor scores. Participants with high blood concentrations of both toxicants had the highest tremor scores, suggesting an additive effect of these toxicants on tremor severity. Given the very high population prevalence of tremor disorders, identifying environmental determinants is important for primary disease prevention.